# 2020 Winter Qual

## III.

(a) $X^TX\beta = X^Ty \Rightarrow X^TX(X^TX)^-X^Ty = X^Ty$

$(X^TX)^-X^Ty =: x_p$ ; particular solution

Note that $\hat{\beta}_h = x_p + x_h$

where $x_h$ is homogeneous solution s.t $X^TXx_h = 0$.

$X^TX(I - (X^TX)^-X^TX) = 0$

So, $(I - (X^TX)^-X^TX)h = x_h$

$\hat{\beta}_h = x_p + x_h$

$= (X^TX)^-X^Ty + (I - (X^TX)^-X^Ty)h \quad , h \in \mathbb{R}^p$

$\boxtimes$

(b) Let $\tilde{\beta} = AY$ s.t LUE.

$E[\tilde{\beta}] = AX\beta = \beta \Rightarrow AX = I$.

$\underset{\underset{p}{\parallel}}{\text{rank}(I)} = \text{rank}(AX) \leq \text{rank}(X) = r$ $\ast$

$\therefore \nexists$ LUE

$\boxtimes$

(c) $e = y - X\hat{\beta}_h$

$$= y - X(X^TX)^-X^Ty - \overbrace{X(I - (X^TX)^-X^TX)h}^{=0}$$

$$= (I - X(X^TX)^-X^T)y$$

$$e^Te = y^T(I - X(X^TX)^-X^T)y \quad \because I - X(X^TX)^-X^T : \text{idempotent}$$

$$tr(I - X(X^TX)^-X^T) = n - r \quad \because r(X(X^TX)^-X^T) = r(X) = r$$

$$\frac{1}{2}\mu^TA\mu = 0$$

$$\frac{e^Te}{\sigma^2} \sim \chi^2(n-r)$$

$$\frac{e^Te}{\sigma^2} = \chi^2_{n-r} \Rightarrow \frac{e^Te}{\chi^2_{n-r}} = \sigma^2$$

$$\sigma^2 \in \left[ \frac{e^Te}{\chi^2_{n-r,1-\alpha/2}}, \frac{e^Te}{\chi^2_{n-r,\alpha/2}} \right]$$

$\blacksquare$

2.

(a) $\widehat{se}(\hat{\beta}) = s\sqrt{v_{jj}}$, $v_{jj}$ = diagonal of $(X^TX)^{-1}$

$$T_j = \frac{\dfrac{\hat{\beta}_j - \beta_{0j}}{\sigma\sqrt{v_{jj}}} \sim N(0,1)}{\sqrt{\dfrac{S^2}{\sigma^2}} \sim \chi^2(n-p)} \perp\!\!\!\perp \sim t_{n-p}$$

$$(X^TX)^{-1}X^T(I - H_X) = 0$$

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(b) $\mathbb{P}(|T_j| > t_{\alpha/2p,\, n-p}) = \frac{\alpha}{p}$

$$\mathbb{P}\left(\bigcup_{j=1}^{p} \{|T_j| > t_{\alpha/2p,\, n-p}\}\right)$$

$$\leq \sum_{j=1}^{p} \mathbb{P}(|T_j| > t_{\alpha/2p,\, n-p})$$

$$= \frac{\alpha}{p} \cdot p = \alpha$$

$\boxtimes$

(c) $S = BB^T \Rightarrow S^{-1} = (B^T)^{-1}B^{-1}$

$v = B^{-1}(\hat{\beta} - \beta_0)$

$u = B^T e_j$

$$T_j^2 = \frac{(e_j(\hat{\beta} - \beta_0))^2}{e_j^T S e_j} \underset{\substack{\uparrow \\ \text{Cauchy} \\ \text{-Schwarz}}}{\leq} (\hat{\beta} - \beta_0)^T S^{-1} (\hat{\beta} - \beta_0)$$

The equality holds when $u \propto v$

$$\Rightarrow B^T e_j \propto B^{-1}(\hat{\beta} - \beta_0)$$

$$\Rightarrow e_j \propto S^{-1}(\hat{\beta} - \beta_0)$$

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(d)

$$T_j^2 \leq (\hat{\beta} - \beta_0)^T \underbrace{S^{-1}}_{= S^2(X^TX)} (\hat{\beta} - \beta_0)$$

$$= \frac{(\hat{\beta} - \beta_0)^T X^T X (\hat{\beta} - \beta_0)}{S^2}$$

$$\Rightarrow \frac{(y - X\beta_0)^T H_X (y - X\beta_0)/p}{(y - X\beta_0)^T (I - H_X)(y - X\beta_0)} \sim F_{p, n-p}$$

$$H_X(I - H_X) = 0$$

$$\Rightarrow (\hat{\beta} - \beta_0)^T S^{-1} (\hat{\beta} - \beta_0) \leq p F_{p, n-p}$$

$$\mathbb{P}\left((\hat{\beta} - \beta_0)^T S^{-1} (\hat{\beta} - \beta_0) \leq p F_{p, n-p}\right) = 1 - \alpha$$

$$\mathbb{P}\left(\bigcap_{j=1}^{p} \{ T_j^2 \leq p F_{p, n-p} \}\right)$$

$$\geq \mathbb{P}\left((\hat{\beta} - \beta_0) S^{-1} (\hat{\beta} - \beta_0) \leq p F_{p, n-p}\right)$$

$$= 1 - \alpha$$

$\boxtimes$